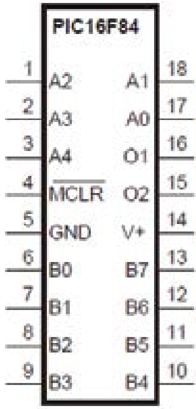
from the microcontroller's memory and executes them. While doing that, it can store data temporarily in RAM (Random Access Memory) and retrieve it. The general purpose working RAM in a PIC is also known as "file registers."

The program memory of the 16F84 is of flash type which means that it can be programmed (stored with data) and erased electrically. So, you can store data in your PIC 16F84, erase it and then restore the data any number of times you want. Also, the memory is non-volatile, which means it retains its contents even when powered off.



Pin diagram of PIC16F84

Every processing unit has number of ports, which essentially is a group of terminals (or pins) used to transfer data. Generally, a port has eight pins. The PIC16F8X has two ports, PORTA and PORTB. Now, these ports are Input / Output Ports, which means they can work as both the input and the output for the data. Moreover, each pin is independent and you can even make two different pins of a same port do different tasks that is one working as input and the other as output. Every port has a register associated with it called a TRIS register. The purpose of TRIS register is to define which pin of that particular port will work as an input and which will work as the output. The TRIS register of PORTA is called TRISA and TRIS register of PORTB is called TRISB. A TRIS register consists of as many bits as there are Input/ Output pins in that port. Assigning the value '1' to a TRISA bit will make the corresponding pin work as an input, while assigning vale '0' to the bit will make the corresponding pin work as an output.

Each output pin of 16F84 can source or sink 20 mA as long as only one pin is doing so at a particular time. For more information on PIC 16F84, you can refer to its datasheet.

5.1.2 Power requirements

According to datasheet of 16F84, it requires 5V power supply, but it can practically work on any voltage between 4.5V to 6V.

Although the PIC consumes only 1 mA or even less at low clock speeds – but the power supply must also provide the current flowing through LEDs or other high-current devices that the PIC may be driving.

The circuit also has a 0.1-μF capacitor connected from pin 14 to ground,